

Investigating the Effects of Display Layouts on Text Readability in Virtual Reality

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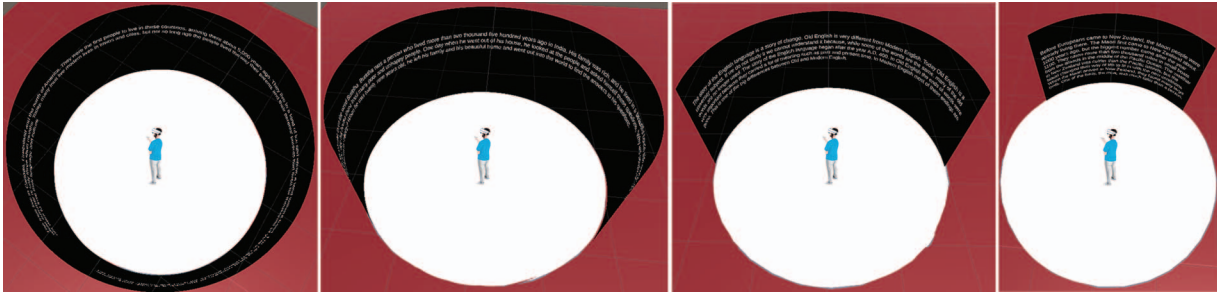
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Figure 1: 4 different Display Layouts for a 2000R curved display: (a) Curve360, (b) Curve270, (c) Curve180 and (d) Curve90. Here, "Curve" refers to the display type; curved numbers such as 360 and 90 refer to the viewing angle of the display.

ABSTRACT

Curved displays, featuring layouts such as semi-circular and fully circular designs, offer superior visual experiences compared to flat screens due to their alignment with human vision. This study assesses user experience in VR with different display layouts, comparing curved designs (quarter circle, semicircle, three-fourths circle, full circle) and flat screens. It focuses on reading comprehension across 8 display types. The study finds that shorter display layouts are preferred for their faster reading speeds compared to longer curved and flat displays. The results offer key insights for designing and optimizing curved displays in virtual reality applications.

Keywords: Display Layout, Curved Display, Text Readability

1 INTRODUCTION

Large displays primarily come in two forms: Flat and Curved. Flat screens, due to their widespread use and familiar viewing experience, have been extensively researched across various domains. Large curved displays enhance user experience by providing detailed information, improving task efficiency, and enhancing satisfaction [1]. Despite the usefulness of curved displays [6], the development of large curved screens is difficult due to hardware and space requirements [2]. Studies [7] show that virtual curved displays perform similarly to physical ones, indicating that virtual environments provide a practical option for exploring different display features without the limitations of space and cost associated with physical displays. Liu et al. compared three display layouts (semi-circle, full-circle, and flat) to be influencing spatial memory [5] and graph comparison [4]. However, there's room to explore other layouts such as the quarter-circle and three-fourths circle. Furthermore, the curved displays that they considered did not have the same display radius (i.e., curvature). Researchers have reported that 2000R performs better than other display curvatures in terms of usability and visual fatigue [3]. Previous research has also suggested considering a viewing distance

equal to the display curvature radius [6]. Despite these advances, comprehensive assessments of text readability across various curved display layouts remain limited.

In this paper, we explore the effect of various curved *Display Layouts* and *Display Types* on text readability within a VR environment. We consider two *Display Types*: Flat and Curved. We have defined four curved *Display Layouts*: Curve90 (quarter circle with a 90-degree viewing angle), Curve180 (semicircle with a 180-degree viewing angle), Curve270 (three-fourths circle with a 270-degree viewing angle), and Curve360 (full circle with a 360-degree viewing angle). Corresponding to each curved layout, we have designed flat displays with equivalent widths. For example, the width of Flat90 matches that of Curve90. We term a combination of curved and flat displays with the same width as a *Display Layout Pair*. In total, we have four Display Layout Pairs: Curve90 and Flat90; Curve180 and Flat 180; Curve270 and Flat270; Curve360 and Flat360. Our key findings reveal that participants were faster in the reading comprehension task with the *Display Layout Pair* 90, 180 and 270 compared to the *Display Layout Pair* 360.

2 USER STUDY: READING COMPREHENSION

In this study, we investigated how display layout and type (flat or curved) affect reading comprehension speeds and accuracy in VR from a corpus of standardized English reading assessments. The interaction distance was set to 2m, the display curvature was 2000R and the text size was 100 pt. We used 20 paragraphs ranging in length from 100 to 110 words. We recruited 16 participants (8 male, 8 female) aged between 20 and 34 years (avg. 26.18, SD 4.58).

3 STUDY RESULTS

We analyze reading time and reading accuracy with repeated measures ANOVA, and pairwise comparisons with Bonferroni corrections. When there is a breach of sphericity, we report p-values and degrees of freedom adjusted using the Greenhouse-Geisser correction.

3.1 Reading time

In terms of reading time, we found significant main effects for the independent variables *display layout pair* ($F_{3,45} = 9.26, p < 0.001, \eta^2 = 0.38$) (see Figure 2a). However, *display type* ($F_{1,15} = 0.41,$

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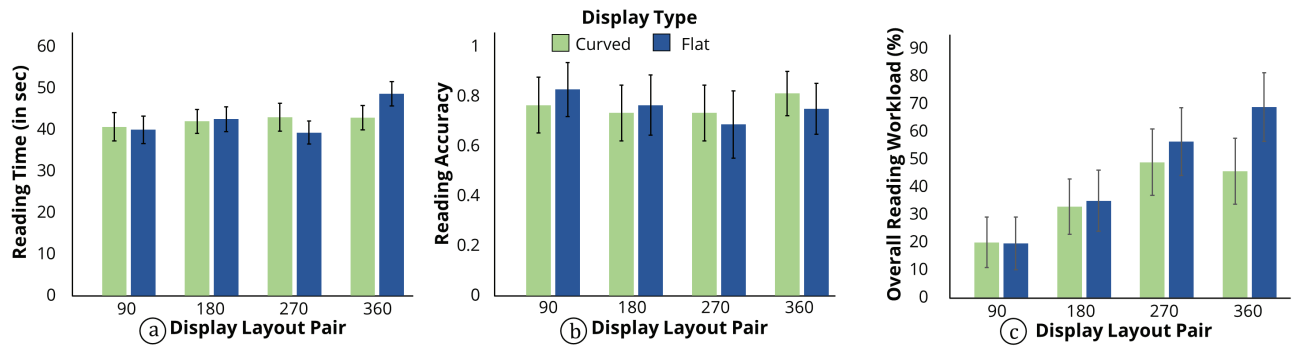


Figure 2: For each *display type* (a) Reading time by *display layout pair* (b) Reading accuracy by *display layout pair* (c) Overall reading workload by *display layout pair*

$p = 0.53$) did not affect on reading time. We found significant interactions of *display type* $\times$ *display layout pair* ($F_{3,45} = 6.51$, $p < 0.001$, $\eta^2 = 0.30$). From pairwise comparisons, we found Curve90, Flat90, and Flat270 are significantly faster (all $p < 0.05$) than Flat360. We found that Curve360 is significantly faster than Flat360 whereas Flat270 is significantly faster than Curve270. No other significant effect was found.

3.2 Reading Accuracy

In terms of accuracy, *display layout pair* ($F_{3,45} = 0.74$, $p = 0.54$) and *display type* ($F_{1,15} = 0.01$, $p = 0.92$) did not affect on reading time (see Figure 2b). Furthermore, we did not find any interactions of *display type* $\times$ *display layout pair* ($F_{3,15} = 0.77$, $p = 0.52$).

3.3 Preference Score

The NASA Task Load Index questionnaire was used to gather user feedback on their mental demand, physical demand, performance, effort, frustration, temporal demand, and overall task load for each of the 8 displays for the reading comprehension task. Additionally, we collected participants' preference and fatigue while reading on each of the displays (*display type* $\times$ *display layout pair*). A Friedman test on the reading comprehension data revealed significant differences were found for *mental demand* ($\chi^2(7, N = 16) = 63.01$, $p < 0.001$), *physical demand* ($\chi^2(7, N = 16) = 55.75$, $p < 0.001$), *performance* ($\chi^2(7, N = 16) = 70.17$, $p < 0.001$), *effort* ($\chi^2(7, N = 16) = 66.561$, $p < 0.001$), *frustration* ($\chi^2(7, N = 16) = 68.74$, $p < 0.001$), *fatigue* ($\chi^2(7, N = 16) = 54.93$, $p < 0.001$), and *overall workload* ($\chi^2(7, N = 16) = 66.57$, $p < 0.001$) (see Figure 2c). However, no significant difference was observed for *temporal demand* ($\chi^2(7, N = 16) = 12.391$, $p = 0.09$). Flat90 had lower *overall workload* than Flat270, Curve270, Flat360, and Curve180. Additionally, Curve90 is better than Curve270 and Flat360 in terms of *overall workload*.

4 DISCUSSION

For the reading task, we observed that the *display type* (whether flat or curved) did not significantly influence both reading time and reading accuracy. This suggests that the inherent advantages or disadvantages of either being flat or curved might not affect reading tasks. However, the subjective feedback from the NASA TLX questionnaire indicates that certain flat displays, especially those with wider display layouts (i.e., Flat360 and Flat270), are more mentally and physically demanding for users. This could be attributed to the increased effort required to navigate and process information on larger displays. Notably, displays like Flat90, Curved90 were preferred over others, indicating that users find certain displays more comfortable and efficient for reading tasks.

Our results highlight the importance of display layout in influencing reading performance. Specifically, participants exhibited faster reading speeds for the *display layout pair* 90 compared to the *display layout pair* 360. This could be attributed to the ergonomic advantages of shorter display layouts, where users can easily view and process information without significant head or eye movement. The feedback from participants, such as "requires no body movement" and "the entire text visible at a glance," further supports this observation. Additionally, the NASA TLX results show that displays like Flat360 were found to be more mentally demanding and effort-intensive compared to others like Curved90 or Flat90, emphasizing the importance of considering user workload in display design based on display layouts.

5 CONCLUSION

This paper examines how display layout and type (flat or curved) in VR affect reading comprehension. We found that shorter displays enhance reading speed. We also gathered user feedback on workload for each display. By using findings, our research offers VR designers guidelines and tips on using display characteristics to boost performance on large virtual displays.

REFERENCES

- [1] C. Ardito, P. Buono, M. F. Costabile, and G. Desolda. Interaction with large displays: A survey. *ACM Comput. Surv.*, 47(3), feb 2015. doi: 10.1145/2682623
- [2] M. Cavallo, M. Dholakia, M. Havlena, K. Ocheltree, and M. Podlasek. Dataspace: A reconfigurable hybrid reality environment for collaborative information analysis. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 145–153, 2019.
- [3] K. Lim, X. Li, and Y. Tu. Effects of curved computer displays on visual performance, visual fatigue, and emotional image quality. *Journal of the Society for Information Display*, 27(9):543–554, 2019.
- [4] J. Liu, A. Prouzeau, B. Ens, and T. Dwyer. Design and evaluation of interactive small multiples data visualisation in immersive spaces. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 588–597, 2020.
- [5] J. Liu, A. Prouzeau, B. Ens, and T. Dwyer. Effects of display layout on spatial memory for immersive environments. *Proc. ACM Hum.-Comput. Interact.*, 6(1SS), nov 2022.
- [6] S. Park, G. Kyung, D. Choi, J. Yi, S. Lee, B. Choi, and S. Lee. Effects of display curvature and task duration on proofreading performance, visual discomfort, visual fatigue, mental workload, and user satisfaction. *Applied ergonomics*, 78:26–36, 2019.
- [7] A. K. M. A. Ullah, W. Delamare, and K. Hasan. Exploring users' pointing performance on virtual and physical large curved displays. In *Proceedings of the 29th ACM Symposium on Virtual Reality Software and Technology*, pp. 1–11, 2023.